Neffke, Otto & Weyh (2017)

The research problem addressed in the paper is that although labor mobility plays an important role in economics, the mobility patterns of workers across industry boundaries and the understanding of inter-industry structure of labor flows has been neglected.

Earlier relatedness measures (such as co-exports, co-occurrence and input-output) often failed to capture the distance between industries in terms of human capital requirements and neglected the inter-industry constraints to labor mobility.

The authors sought to answer how industry-specific human capital is and which industries have similar human capital requirements. Their main contribution is the development of a novel labor-flow-based skill-relatedness index that significantly outperforms previous co-location and input-output measures in predicting the entry and regional growth of local industries. Furthermore, they contributed to the understanding of inter-industry labor flows by showing that these flows are highly concentrated among a few industry pairs and form a network that is independent of a worker's occupation and geographic mobility (local or large distance).

Penny Mealy, Alexander Teytelboym (2022) – Economic complexity and the green economy

They have extended Neffke et al. into a new context and data by applying capability-based relatedness models to the green economy. They used a dataset of traded green products and demonstrated how countries leverage existing productive capabilities for green diversification and carbon emissions reduction, which is also a new outcome.

R. Maria del Rio-Chanona et al. (2021) – Occupational mobility and automation: a data-driven network model

They have extended Neffke et al. by applying occupational mobility networks to analyze worker reallocation under automation shocks which gives a new context and mechanism to Neffke et al's work. They developed a data-driven network model using occupational transition data to simulate macro-level labor market dynamics and micro-level unemployment outcomes across different wage groups, which refers to new method and new outcome.

My own niche on skill relatedness: how university students choose their master's study fields and how can they succeed based on their previous studies (i.e based on skills acquired during previous studies)

Research question: How does the degree of skill-relatedness between undergraduate and postgraduate fields affect students' chocies when selecting a master's program? To what extent does a higher skill-relatedness predict academic success and program completion in master's studies?

Previous work has shown that skill-relatedness drives worker mobility across industries (Neffke et al., 2017) but we do not know how skill-relatedness between undergraduate and postgraduate disciplines influences student's choices of master's programs and their subsequent academic performance. This matters because higher education institutions need data-driven insights to design flexible interdisiplinary curricula, optimize admission requirements and reduce drop-out rates.

Data source to answer the question:

- student transition records tracking student admissions and enrolment flows from specific Bachelor's degree programs to master's programs
- Provided and required skills by programs: mapping the skills provided by undergraduate curricula and required by master's programs (ex: extracting data from syllabus)
- academic performance metrics, ex: course grades, program completion rates, overall grade average (grade point average)

Who would use the answer: university leaders and academic decision makers, program directors, educational policy makers